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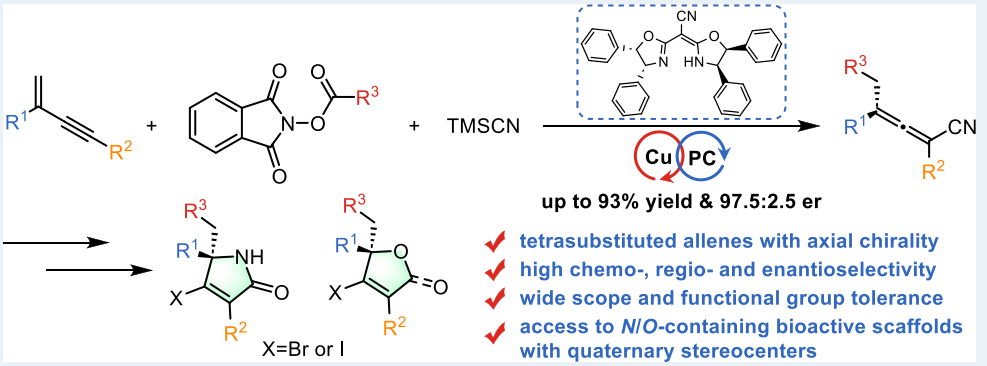
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Metallaphotoredox Synthesis of Axially Chiral Tetrasubstituted Allenes through Regio- and Enantioselective 1,4-Carbocyanation of 1,3-Enynes

Keyi Peng, Ya-Sheng Cao, Quanyuan Wang, Ziwei Xia, Ya Chen,* Yixin Lu,* and Guo-Jun Deng*

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ABSTRACT: Radical 1,4-difunctionalization of 1,3-enynes represents a promising approach for the rapid creation of allenes from readily available feedstocks. However, the enantioselective synthesis of tetra-substituted chiral allenes from 1,3-enynes via a radical pathway remains difficult. Herein, we disclose an enantioselective decarboxylative 1,4-carbocyanation of 1,3-enynes through photoredox and copper dual catalysis, affording a wide array of axially chiral tetra-substituted allenes in a highly chemoselective, regioselective, and enantioselective manner. This mild and redox-neutral protocol exhibits broad substrate scope and good functional group tolerance and enables late-stage allenyation of natural and drug-like molecules.

KEYWORDS: *asymmetric radical reaction, metallaphotoredox, axial chirality, tetrasubstituted allenes, 1,3-enynes*

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1. INTRODUCTION

Due to the high reactivity of open-shell intermediates, catalytic asymmetric photochemical transformations present a significant synthetic challenge.[1] In recent years, the dual catalysis strategy, which combines photoredox catalysis with organic catalysis,[2] transition-metal catalysis,[3] or enzyme catalysis,[4] has offered an efficient solution for constructing chiral molecules. Nevertheless, the synthesis of axially chiral compounds[5] via photo-induced processes is still in its early stages, with limited examples reported (Scheme 1a). In their pioneering work, Bach and co-workers reported a catalytic deracemization of allene lactams using a chiral bifunctional thioxanthone photosensitizer via a triplet energy-transfer process.[6] Wu et al. disclosed a photo-induced asymmetric synthesis of axially chiral styrenes, through SO_2 insertion and utilizing derived cinchona alkaloids as an organocatalyst.[7] The Xiao group developed an asymmetric Minisci reaction for constructing axially chiral heterobiaryls by combining photoredox and chiral phosphoric acid catalysis.[8] The same group also achieved the formation of various (hetero)biaryls with axial chirality through the cooperative photoredox and cobalt catalysis.[9] Subsequently, Li and co-workers extended the system to the synthesis of axially and centrally chiral diaryl ethers.[10] Furthermore, the Wang group developed a photoredox and chromium cooperative catalysis to prepare trisubstituted chiral α-allenols.[11] Despite these impressive advances, the exploration of novel multiple catalysis systems for the enantioselective construction of structurally diverse axial chiral skeletons remains intellectually challenging and interesting.

Axially chiral allenes are widely present in natural products, therapeutic agents, and functional materials, and they are also tremendously useful as intermediates in organic synthesis.[12] Therefore, searching for efficient asymmetric catalytic approaches to access these molecules has attracted much attention from synthetic chemists.[13] Given the ready availability of 1,3-enynes, their asymmetric 1,4-difunctionalization offers a highly attractive method for the synthesis of chiral allenes.[14] In this

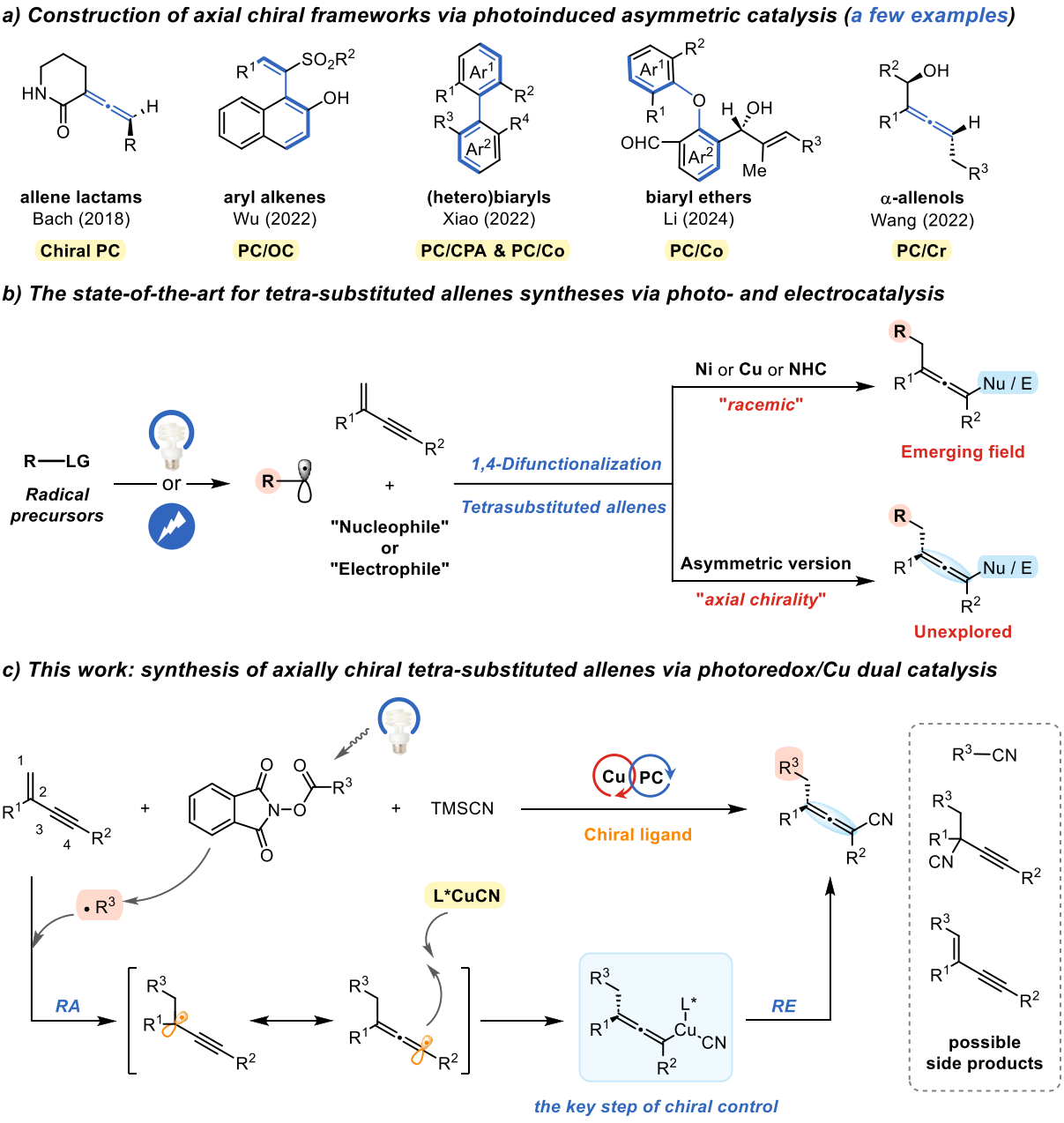
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Scheme 1. Photoinduced Construction of Axially Chiral Frameworks

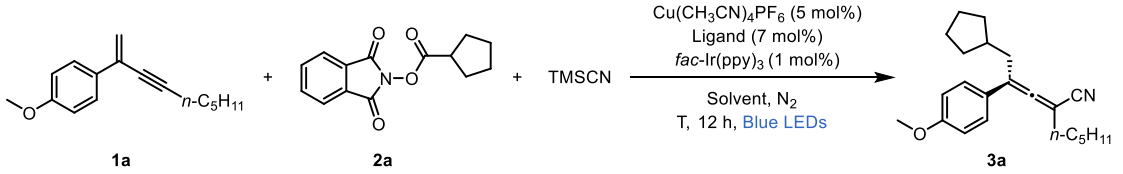


context, most reported methods are focused on the synthesis of optically enriched di- or trisubstituted allenes,[15] and only a few examples are applicable to the synthesis of challenging tetra-substituted chiral allenes via transition-metal catalysis,[16] or organocatalysis.[17] Recently, 1,4-difunctionalization of 1,3-enynes via radical pathways emerged as a powerful approach for the synthesis of tetra-substituted allenes.[18] Nevertheless, when the enantioselective synthesis is concerned, achieving good stereochemical control of allenyl radicals is a very challenging task due to their linear geometry.[19] Very recently, in their elegant studies of copper-catalyzed asymmetric radical 1,4-difunctionalization of 1,3-enynes, Bao[20] and Liu[21] made use of radical species that are generated from peroxides and electron-deficient alkyl halides, respectively. Given the widespread applications of photoredox catalysis in organic synthesis and the ready availability of various radical precursors, an enantioselective synthesis of tetra-substituted allenes involving a photoredox pathway would be highly desirable. It was with this goal in mind that we initiated this study.

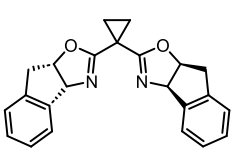
Utilizing dual photoredox and transition-metal catalysis, we introduced a new approach for site-selective radical difunctionalization of 1,3-enynes.[22] It is noteworthy that this research area sees explosive development; numerous strategies relying on the combination of photo- and electrochemistry with transition-metal catalysis or organocatalysis have emerged for radical 1,4-difunctionalization of 1,3-enynes, offering efficient synthesis of structurally diverse allene products from readily available feedstock (Scheme 1b).[23−25] However, enantioselective 1,4-difunctionalization remains a difficult task. Given our continuous interest in asymmetric photochemical transformations[26] and construction of significant axially chiral scaffolds,[27] we envisioned that a metallaphotoredox system could be applied to achieve enantiocontrol over highly reactive allenyl radicals, leading to the formation of tetra-substituted chiral allenes. The

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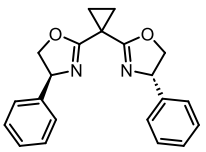
Table 1. Optimization of Reaction Conditions[a]



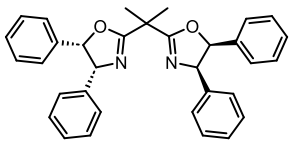
1a 2a 3a



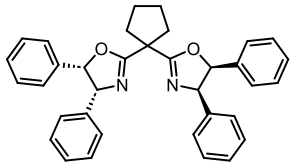
L1, 40%, 57.5:42.5 er



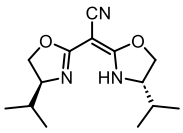
L2, 61%, 34.5:65.5 er



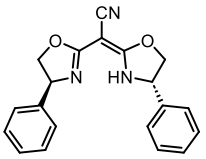
L3, 31%, 61:39 er



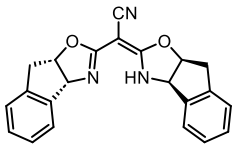
L4, 22%, 60:40 er



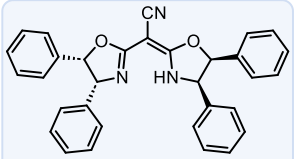
L5, 66%, 12:88 er



L6, 47%, 12.5:87.5 er



L7, 57%, 82.5:17.5 er



L8, 85%, 91.5:8.5 er

Entry	Ligand	T (°C)	Solvent	Yield (%)[b]	er[c]	Entry	Ligand	T (°C)	Solvent	Yield (%)[b]	er[c]
1	**L8**	45	DMAc	85	91.5:8.5	6	**L8**	30	DMPU	76	93:7
2	**L8**	45	DMSO	74	87.5:12.5	7	**L8**	0	DMPU	40	95:5
3	**L8**	45	CH_3CN	38	65.5:34.5	8[d]	**L8**	45	DMPU	88	96.5:3.5
4	**L8**	45	DMPU	92	93.5:6.5	9[d,e]	**L8**	45	DMPU	86	96.5:3.5
5	**L8**	60	DMPU	94	93:7	10[d,e]	w/o Cu cat., ligand, PC or light			0	---

[a]Reaction conditions: **1a** (0.2 mmol), **2a** (0.2 mmol), TMSCN (0.1 mmol), $Cu(CH_3CN)_4PF_6$ (5 mol %), ligand (7 mol %), *fac*-Ir(ppy)$_3$ (1 mol %), solvent (1.0 mL), 45 °C, N_2 atmosphere, 35 W Blue LEDs, 12 h. [b]Isolated yield. [c]The er values were determined by HPLC analysis on a chiral stationary phase. [d]*Fac*-Ir(ppy)$_3$ (0.15 mol %). [e]**1a** (0.15 mmol), **2a** (0.15 mmol).

key in our designed process is the stereoselective trapping of photoredox-generated allenyl radicals by engaging a suitable chiral transition-metal complex. Meanwhile, we are mindful of the inherent difficulty in circumventing the undesired two-component coupling and 1,2-addition to 1,3-enynes.[28] Herein, we disclose an enantioselective synthesis of tetra-substituted allenes via decarboxylative 1,4-carbocyanation of 1,3-enynes under cooperative photoredox and copper catalysis (Scheme 1c).

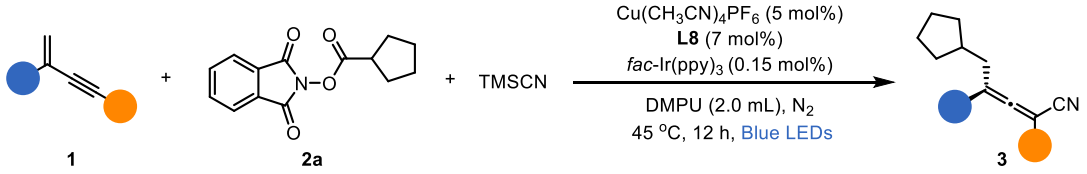
2. RESULTS AND DISCUSSION

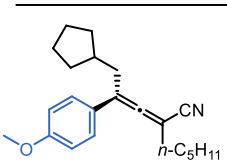
We initiated our investigation by choosing 1,3-enyne **1a** as the model substrate, *N*-hydroxyphthalimide (NHPI) ester **2a** as the radical precursor, and TMSCN as the nucleophilic reagent, and the results are summarized in Table 1. A series of commercially available chiral bisoxazoline (Box) ligands were first examined (for details, see the Supporting Information), in the presence of 5 mol % $Cu(CH_3CN)_4PF_6$ and 1 mol % *fac*-Ir(ppy)$_3$ in DMAc under irradiation with blue LEDs at 45 °C. Notably, the common neutral Box ligands that worked well in radical cyanation[29] led to poor asymmetric induction (**L1**−**L4**). In contrast, the anionic cyanobisoxazoline (CN-Box) ligands (**L5**−**L8**) turned out to be effective. The CN-Box **L8** was quickly established as the best ligand, forming the desired chiral tetra-substituted allene in 85% yield and good enantiomeric control (91.5:8.5 er). The subsequent solvent screening was followed, and DMPU was identified to be the solvent of choice, and the enantiomeric ratio was improved to 93.5:6.5 (entries 1−4). Reaction optimization by varying the reaction temperatures was followed (entries 5−7). Even though the running reaction at 0 °C slightly improved the enantioselectivity, the yield of the reaction dropped significantly. Other reaction parameter, such as copper salts, photo photosensitizers, light source, and the ratios of reactants were all carefully evaluated (for details, see the Supporting Information). Finally, lowering the photosensitizer *fac*-Ir(ppy)$_3$ loading to 0.15 mol % further enhanced the er ratio to 96.5:3.5, with virtually maintained high yield (entry 8), which was attributed to the slow racemization of the allene product under 1 mol % photosensitizer (for details, see the Supporting Information). As anticipated, control experiments confirmed that the CN-Box/Cu^I catalyst, photosensitizer, and light were all essential for achieving the desired transformation (entry 10).

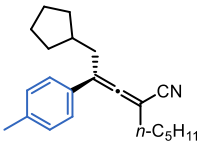
After the optimized reaction conditions were established, the substrate scope with regard to 1,3-enynes was then investigated. As shown in Table 2, various substituents on the phenyl ring that was connected with the double bond of enynes were well-tolerated, forming tetra-substituted allenes in good to excellent yields with excellent enantiomeric controls (**3a**−**3h**). Notably,

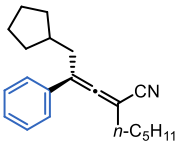
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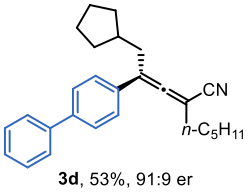
Table 2. Substrate Scope of 1,3-Enynes[a]

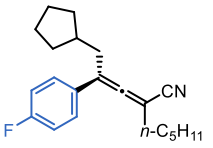


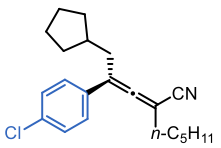

3a, 86%, 96.5:3.5 er

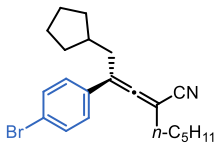

3b, 92%, 94.5:5.5 er

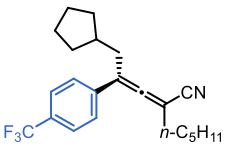

3c, 85%, 95:5 er

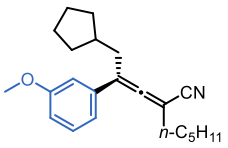

3d, 53%, 91:9 er

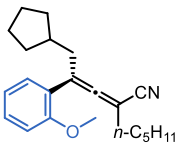

3e, 83%, 93.5:6.5 er

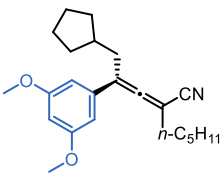

3f, 74%, 94:6 er

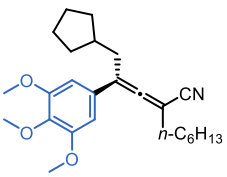

3g, 75%, 94:6 er

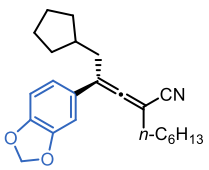

3h, 66%,92.5:7.5 er

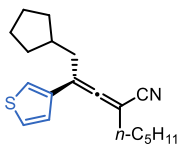

3i, 88%, 96:3 er

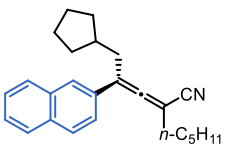

3j, 63%, 90.5:9.5 er

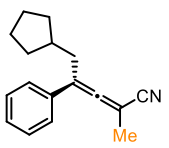

3k, 85%,96:4 er

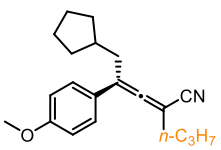

3l, 84%, 97.5:2.5 er

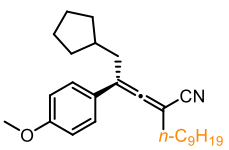

3m, 73%, 90:10 er

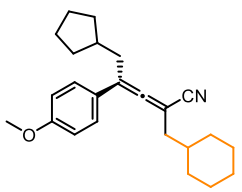

3n,74%, 96:3 er

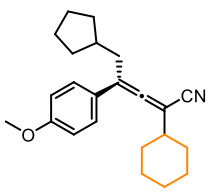

3o, 57%,72.5:27.5 er

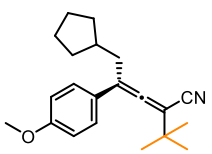

3p, 61%, 94:6 er

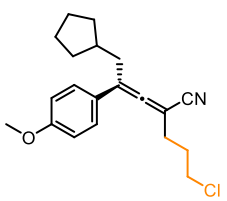

3q, 90%, 95.5:4.5 er

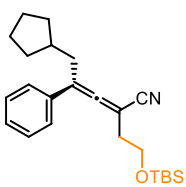

3r, 85%, 95.5:4.5 er

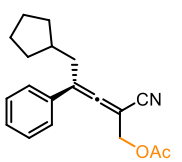

3s, 81%, 95.5:4.5 er

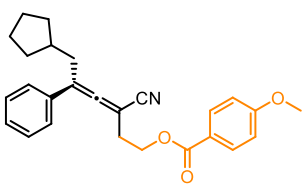

3t, 62%, 93.5:6.5 er


3u, 34%, 86:14 er


3v, 90%,94.5:5.5 er


3w, 72%, 94.5:5.5 er


3x, 58%,92.5:7.5 er


3y, 76%, 92:7 er

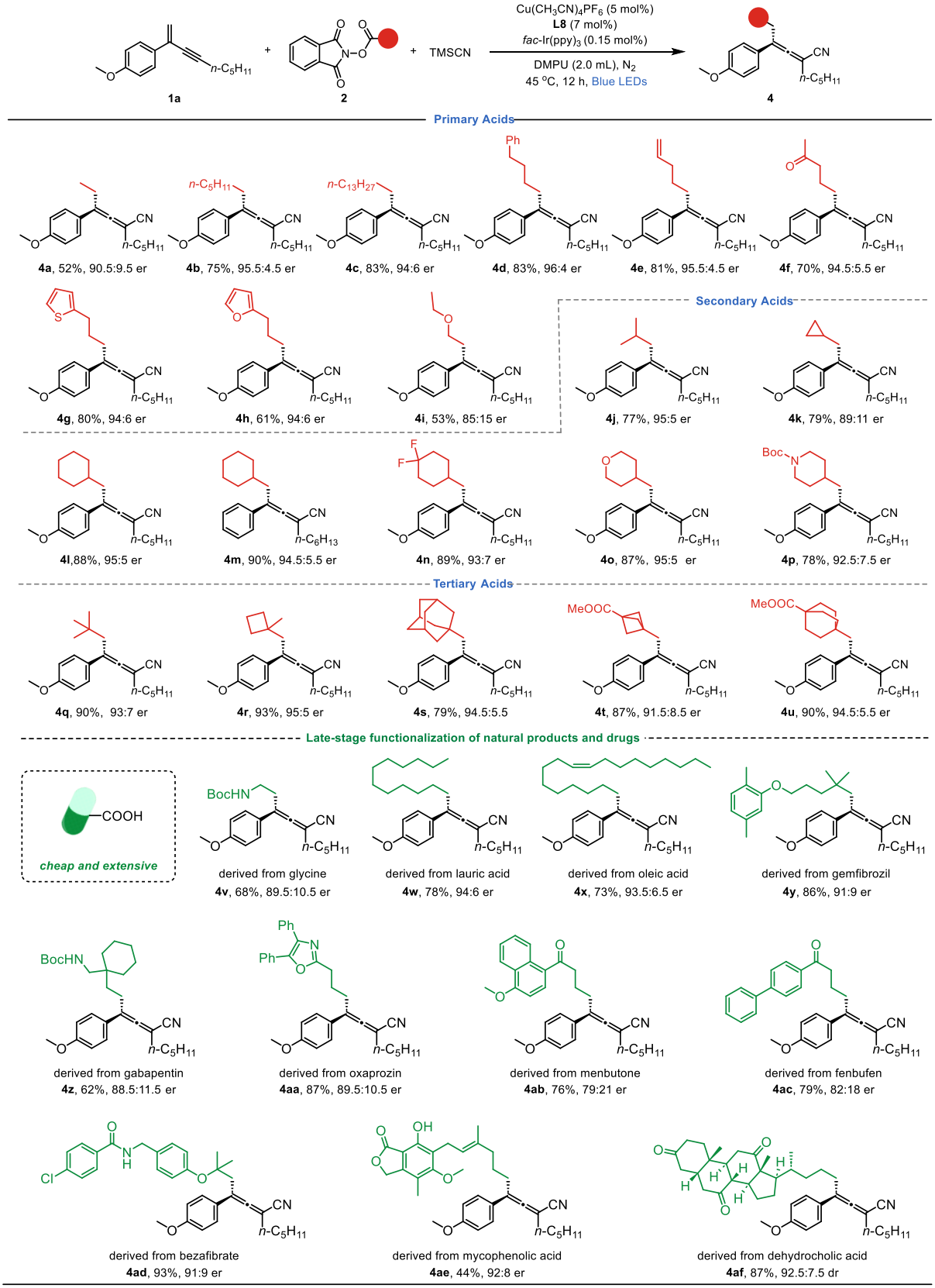
[a]Reaction conditions: **1** (0.3 mmol), **2a** (0.3 mmol), TMSCN (0.2 mmol), $Cu(CH_3CN)_4PF_6$ (5 mol %), **L8** (7 mol %), *fac*-$Ir(ppy)_3$ (0.15 mol %), DMPU (2.0 mL), 45 °C, N_2 atmosphere, 35 W blue LEDs, 12 h. Isolated yield. The er values were determined by HPLC analysis on a chiral stationary phase.

1,3-enynes bearing *para*-, *meta*- and *ortho*-methoxy-substituted phenyls all worked, suggesting that the steric hindrance of the substituents is inconsequential (**3a**, **3i**, and **3j**). 1,3-Enynes with a di- and trisubstituted aryl moiety also proved to be good substrates (**3k** and **3l**). Moreover, enynes containing a heterocycle, e.g., 1,3-benzodioxole, or thiophene, worked equally well for the reaction, and the desired allenes were obtained in high yields with good enantiomeric controls (**3m** and **3n**). When a naphthyl-containing 1,3-enyne was employed, the enantioselectivity of the reaction dropped significantly (**3o**). The alkynyl moieties of 1,3-enynes were next examined. The methyl and linear alkyl substituents of different lengths, as well as the cycloalkyl groups, were all well-tolerated (**3p**−**3s**). When the cyclohexyl group was directly attached with the alkynyl moiety, the desired product was obtained in an acceptable yield and er value (**3t**). However, the employment of 1,3-enyne with a sterically hindered *tert*-butyl group led to a much reduced enantiomeric control (**3u**). Finally, 1,3-enynes with an alkyl chain decorated with different functional groups, e.g., chlorine, silyl ether, and esters, were good substrates as well, and the allene products were obtained in good yields with excellent er values (**3v**−**3y**).

The substrate scope with regard to redox-active alkyl esters was next examined (Table 3). A broad range of NHPI esters **2** derived from primary, secondary, and tertiary alkyl carboxylic acids were employed. Primary alkyl groups of different chain

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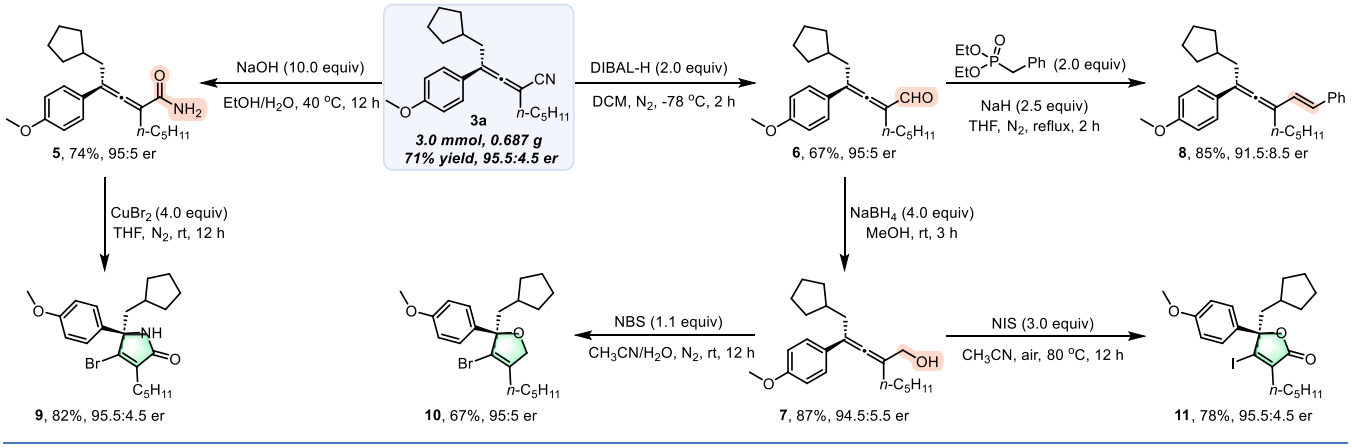
Table 3. Substrate Scope of NHPI Esters[a]



[a]Reaction conditions: **1a** (0.3 mmol), **2** (0.3 mmol), TMSCN (0.2 mmol), $Cu(CH_3CN)_4PF_6$ (5 mol %), **L8** (7 mol %), *fac*-Ir(ppy)$_3$ (0.15 mol %), DMPU (2.0 mL), 45 °C, N_2 atmosphere, 35 W blue LEDs, 12 h. Isolated yield. The er values were determined by HPLC analysis on a chiral stationary phase.

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Scheme 2. Gram-Scale Reactions and Synthetic Applications



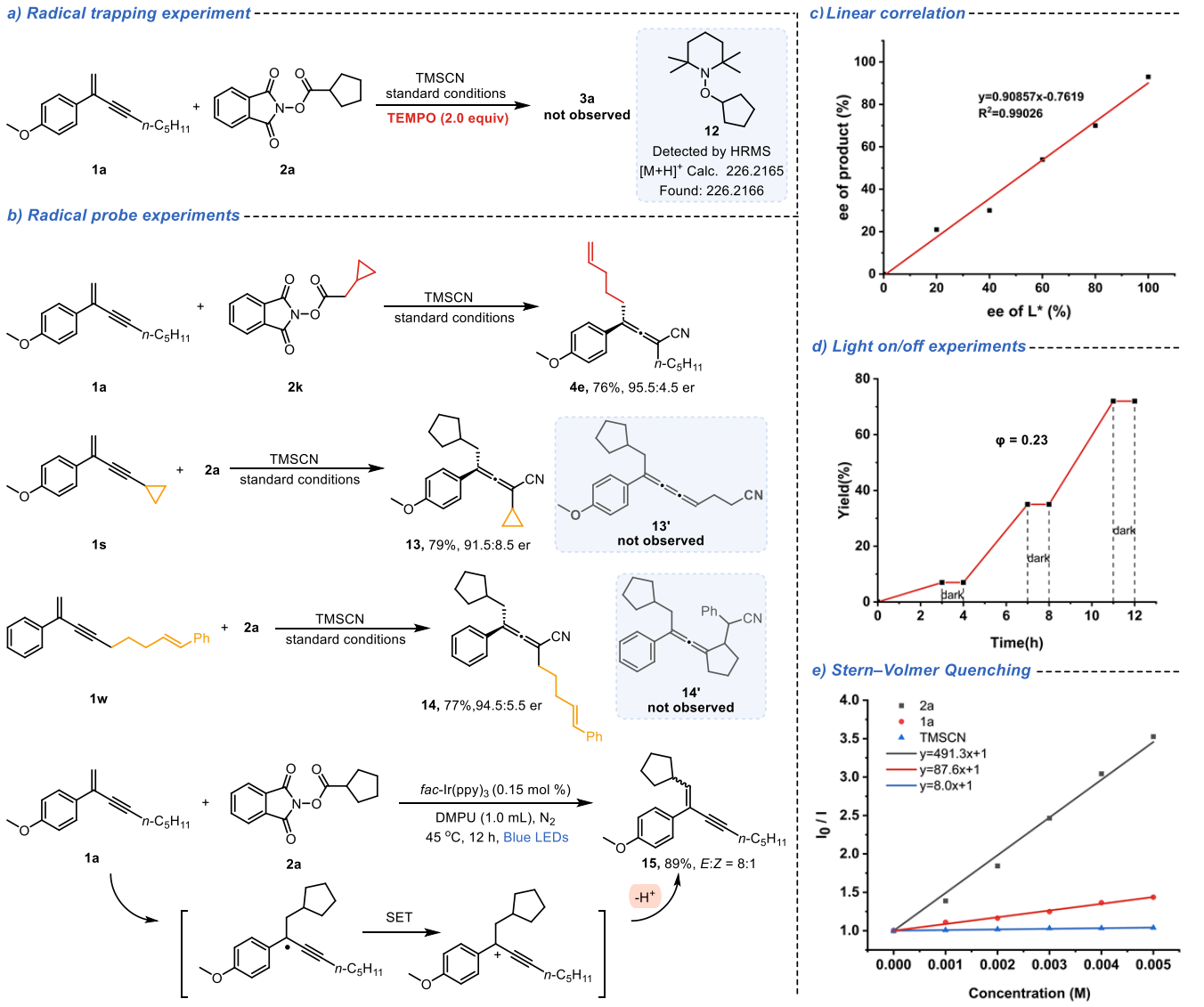


Figure 1. Mechanistic investigation: (a) radical trapping experiment; (b) radical probe experiments; (c) linear correlation between the ee of ligand **L8** and the ee of **3a**; (d) light on/off experiments; and (e) fluorescence-quenching experiments.

lengths (**4b** and **4c**), as well as alkyl substituents containing a phenyl ring (**4d**), an alkene (**4e**), a ketone (**4f**), and heteroarenes (**4g** and **4h**), were found to be compatible, forming the desired allenes in good yields with excellent enantioselectivities. It is interesting to note that even the methyl radical was suitable for the reaction, and the corresponding allene (**4a**) was obtained in moderate yield and good enantioselectivity.

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Secondary alkyl carboxylic acid-derived NHPI esters were also shown to be suitable substrates (**4j**−**4p**). Five- and six-membered cyclic-acid-derived NHPI esters were excellent substrates, forming enantiomerically enriched allene products containing cyclohexyl, pyran, and piperidine moieties. In contrast, the cyclopropyl substrate was less ideal (**4k**). Lastly, a number of highly sterically hindered tertiary alkyl NHPI esters were also investigated, and it is intriguing to notice that the tetra-substituted allenes bearing very sterically demanding groups were obtained in high yields and excellent enantioselectivities (**4q**−**4u**). It is noteworthy that difunctionalization protocol of 1,3-enynes reported herein is regiospecific; we did not observe any 1,2-addition byproducts during the reaction.

Alkyl carboxylic acid structures are widely present in natural products and drug molecules; thus, late-stage allenylation reaction was performed, leading to the generation of complex allene structures (Table 3). Exposure of glycine-derived ester afforded allene **4v** in a good yield. Fatty acids, including lauric acid and oleic acid, were readily converted to the corresponding allene products in good yields with excellent enantioselectivity (**4w** and **4x**). Moreover, alkyl carboxylic acids derived from drug molecules, such as gemfibrozil (**4y**), gabapentin (**4z**), oxaprozin (**4aa**), menbutone (**4ab**), and fenbufen (**4ac**), could all be employed in this asymmetric 1,4-carbocyanation reaction of 1,3-enynes. In particular, the functional group tolerance of this protocol has been fully demonstrated; highly functional drug molecules, e.g., bezafibrate (**4ad**), mycophenolic acid (**4ae**), and dehydrocholic acid (**4af**) smoothly formed the allenylation products.

Synthetic applications of allene products are illustrated (Scheme 2). A gram-scale synthesis of tetra-substituted allene **3a** was performed under the standard reaction conditions. Under basic conditions, allenyl cyanide **3a** was readily hydrolyzed to allenamide **5** without enantiomeric erosion. The selective reduction of **3a** with diisobutylaluminum hydride (DIBAL-H) afforded **6** with a 95:5 er, which could be further transformed to allenyl alcohol **7** and the analogue of insect pheromone **8** via a Wittig reaction. In addition, the transfer of axial chirality to central chirality was investigated. Through copper-mediated cyclizative bromination of **5**, highly functionalization lactam **9** with a quaternary carbon center was obtained in an enantiomerically enriched form. Treating allenyl alcohol **7** with NBS led to the creation of enantioenriched 2,5-dihydrofuran **10**. When **7** was treated with NIS, sequential electrophilic iodinating cyclization and oxidation in the air took place, forming iodinated 2-(5*H*)-furanone **11**. The vinyl bromide and iodide moieties in the above structures can be subjected to further cross-coupling reactions, yielding heteroatom-containing skeletons with promising biological activities.[30]

A series of control experiments were performed to probe the reaction mechanism. With the addition of the radical scavenger TEMPO, the reaction was inhibited. We also detected the TEMPO adduct **12**, indicating the formation of an alkyl radical species (Figure 1a). Several radical clock studies were carried out (Figure 1b). The use of NHPI ester **2k** with a cyclopropyl group afforded exclusively ring-opened allene **4e**, strongly suggesting the intermediacy of alkyl radicals. However, when a 1,3-enyne bearing a ring-opening or ring-closing radical probe at the R^2 position (**1s** and **1w**) was subjected to the standard reactions, the corresponding allenes **13** and **14** were smoothly obtained, and the isomerized products **13′** and **14′** were not observed. These results support that the coupling rate of the allenyl radical with the copper complex is faster than the rate of the radical ring-opening and ring-closing reactions. Treatment of 1,3-enyne **1a** and NHPI ester **2a** with *fac*-Ir(ppy)$_3$ under blue LEDs afforded Heck-type product **15** in 89% yield, indicating the existence of the propargyl radical in the reaction. A linear correlation between the ee values of ligand **L8** and the ee values of product **3a** was observed, suggesting that the active catalytic species is a monomeric copper complex with a single chiral ligand (Figure 1c). The light on/off experiment showed that the continuous irradiation was essential for this reaction, and combined with the quantum yield (φ = 0.23), the radical chain mechanism could be ruled out (Figure 1d). Furthermore, fluorescence-quenching experiments were conducted, and it was found that the excited-state photocatalyst *fac*-Ir(ppy)$_3$ was quenched by NHPI ester **2a**, rather than 1,3-enyne **1a** and TMSCN, supporting an oxidative quenching pathway (Figure 1e).

On the basis of the above mechanistic studies and the literature precedents,[31] a plausible mechanism is depicted in Figure 2. Reduction of NHPI ester **2** ($E_{1/2}$ = −1.26 to −1.37 V vs

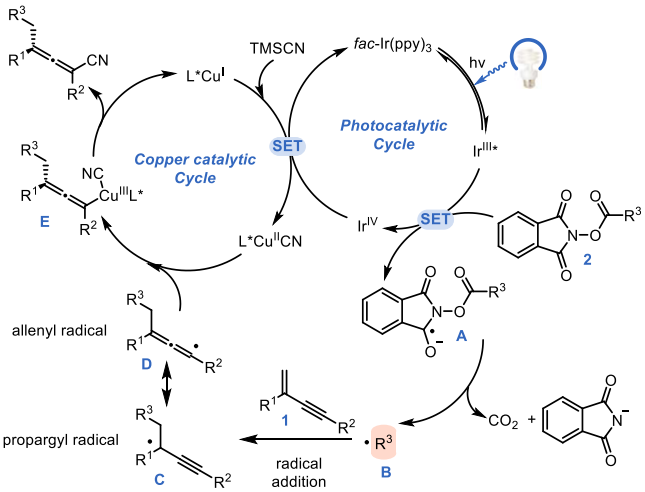


Figure 2. Proposed mechanism.

SCE)[32] by excited-state *fac*-Ir(ppy)$_3$* ($E_{1/2}^{IV/III*}$ = −1.73 V vs SCE, $E_{1/2}^{IV/III}$ = +0.77 V vs SCE)[33] species delivers the oxidative photocatalyst Ir^{IV} and a carboxyl radical anion species **A**, which undergo fragmentation to generate an alkyl radical **B** with extrusion of carbon dioxide and a phthalimide anion. Regioselective addition of the alkyl radical to the double bond of 1,3-enyne **1** affords propargyl radical **C**, which is in resonance with allenyl radical **D**. Meanwhile, L*Cu^{I} is oxidized to L*Cu^{II}CN by Ir^{IV} in the presence of TMSCN. The following stereoselective allenyl radical rebound with the L*Cu^{II}CN species produces a chiral Cu^{III} species **E**. Subsequent reductive elimination furnishes the desired allene product with high enantioselectivity, along with the regeneration of L*Cu^{I} species.

3. CONCLUSIONS

In conclusion, we have developed an enantioselective decarboxylative 1,4-carbocyanation of 1,3-enynes via photoredox and copper cooperative catalysis for the synthesis of axially chiral tetra-substituted allenes. The reaction is conducted under mild reaction conditions, without the use of external oxidants, displaying excellent chemoselectivity, regioselectivity, and enantioselectivity. The scope of the reaction is broad, applicable

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to various primary, secondary, and tertiary NHPI esters. Furthermore, this method is amenable for the late-stage functionalization of natural products and drug-like molecules. We anticipate that this metallaphotoredox dual catalysis will open a new avenue for the asymmetric synthesis of axially chiral allenes and their applications in organic synthesis and medicinal chemistry.

4. METHODS

An oven-dried 10 mL sealed tube containing a stirring bar was charged with 1,3-enyne **1** (0.3 mmol, 1.5 equiv), NHPI esters **2** (0.3 mmol, 1.5 equiv), $Cu(CH_3CN)_4PF_6$ (3.8 mg, 0.01 mmol, 5 mol %), **L8** (6.8 mg, 0.014 mmol, 7 mol %), *fac*-$Ir(ppy)_3$ (0.2 mg, 0.0003 mmol, 0.15 mol %), and DMPU (2.0 mL, 0.1 M) under a nitrogen atmosphere, and then TMSCN (0.2 mmol) was added via a syringe. The resulting mixture was stirred for 12 h under irradiation with blue strip at 45 °C. After completion of the reaction, the resulting mixture was diluted in EtOAc, transferred to a 125 mL separatory funnel, washed with brine, dried over anhydrous $MgSO_4$, and concentrated by rotary evaporation. The residue was purified by column chromatography on silica gel or thin-layer chromatography on silica gel using petroleum ether/ethyl acetate as an eluent to afford the corresponding allene products.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c04330.

Experimental procedure, optimization data, characterization data for all new products, NMR spectra, and HPLC spectra (PDF)

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Author Contributions

Y.C., Y.L., and G.-J.D. directed the project. K.P., Y.-S.C., Q.W., and Z.X. planned and carried out the experiments and prepared the Supporting Information. Y.C., Y.L., G.-J.D., and K.P. co-wrote the manuscript. All authors discussed the results and commented on the manuscript.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) (a) Brimioulle, R.; Lenhart, D.; Maturi, M. M.; Bach, T. Enantioselective catalysis of photochemical reactions. *Angew. Chem., Int. Ed.* **2015**, *54*, 3872−3890. (b) Yoon, T. P. Photochemical stereocontrol using tandem photoredox−chiral lewis acid catalysis. *Acc. Chem. Res.* **2016**, *49*, 2307−2315. (c) Genzink, M. J.; Kidd, J. B.; Swords, W. B.; Yoon, T. P. Chiral photocatalyst structures in asymmetric photochemical synthesis. *Chem. Rev.* **2022**, *122*, 1654−1716.

(2) (a) Silvi, M.; Melchiorre, P. Enhancing the potential of enantioselective organocatalysis with light. *Nature* **2018**, *554*, 41−49. (b) Yin, Y.; Zhao, X.; Qiao, B.; Jiang, Z. Cooperative Photoredox and Chiral Hydrogen-Bonding Catalysis. *Org. Chem. Front.* **2020**, *7*, 1283−1296. (c) Yao, W.; Bazan-Bergamino, E. A.; Ngai, M.-Y. Asymmetric photocatalysis enabled by chiral organocatalysts. *ChemCatChem* **2022**, *14*, No. e202101292.

(3) (a) Lipp, A.; Badir, S. O.; Molander, G. A. Stereoinduction in metallaphotoredox catalysis. *Angew. Chem., Int. Ed.* **2021**, *60*, 1714−1726. (b) Chan, A. Y.; Perry, I. B.; Bissonnette, N. B.; Buksh, B. F.; Edwards, G. A.; Frye, L. I.; Garry, O. L.; Lavagnino, M. N.; Li, B. X.; Liang, Y.; Mao, E.; Millet, A.; Oakley, J. V.; Reed, N. L.; Sakai, H. A.; Seath, C. P.; MacMillan, D. W. C. Metallaphotoredox: The merger of photoredox and transition metal catalysis. *Chem. Rev.* **2022**, *122*, 1485−

1542. (c) Zhang, H.-H.; Chen, H.; Zhu, C.; Yu, S. A review of enantioselective dual transition metal/photoredox catalysis. *Sci. China Chem.* **2020**, *63*, 637−647.
(4) Schmermund, L.; Jurkaš, V.; Özgen, F. F.; Barone, G. D.; Büchsenschütz, H. C.; Winkler, C. K.; Schmidt, S.; Kourist, R.; Kroutil, W. Photo-biocatalysis: biotransformations in the presence of light. *ACS Catal.* **2019**, *9*, 4115−4144.
(5) (a) Cheng, J. K.; Xiang, S.-H.; Li, S.; Ye, L.; Tan, B. Recent advances in catalytic asymmetric construction of atropisomers. *Chem. Rev.* **2021**, *121*, 4805−4902. (b) Zhang, Z.; Dai, L. Construction of axially chiral molecules enabled by photoinduced enantioselective reactions. *Chem. Sci.* **2024**, *15*, 12636−12643.
(6) (a) Hölzl-Hobmeier, A.; Bauer, A.; Silva, A. V.; Huber, S. M.; Bannwarth, C.; Bach, T. Catalytic deracemization of chiral allenes by sensitized excitation with visible light. *Nature* **2018**, *564*, 240. (b) Plaza, M.; Jandl, C.; Bach, T. Photochemical deracemization of allenes and subsequent chirality transfer. *Angew. Chem., Int. Ed.* **2020**, *59*, 12785−12788.
(7) Zhang, C.; Tang, Z.; Qiu, Y.; Tang, J.; Ye, S.; Li, Z.; Wu, J. Access to axially chiral styrenes via a photoinduced asymmetric radical reaction involving a sulfur dioxide insertion. *Chem Catal.* **2022**, *2*, 164−177.
(8) Liang, D.; Chen, J.-R.; Tan, L.-P.; He, Z.-W.; Xiao, W.-J. Catalytic asymmetric construction of axially and centrally chiral heterobiaryls by minisci reaction. *J. Am. Chem. Soc.* **2022**, *144*, 6040−6049.
(9) (a) Jiang, X.; Xiong, W.; Deng, S.; Lu, F.-D.; Jia, Y.; Yang, Q.; Xue, L.-Y.; Qi, X.; Tunge, J. A.; Lu, L.-Q.; Xiao, W.-J. Construction of axial chirality via asymmetric radical trapping by cobalt under visible light. *Nat. Catal.* **2022**, *5*, 788−797. (b) Xiong, W.; Jiang, X.; Wang, W.-C.; Cheng, Y.; Lu, L.-Q.; Gao, K.; Xiao, W.-J. Dynamic kinetic reductive conjugate addition for construction of axial chirality enabled by synergistic photoredox/cobalt catalysis. *J. Am. Chem. Soc.* **2023**, *145*, 7983−7991. (c) Jiang, H.; He, X.-K.; Jiang, X.; Zhao, W.; Lu, L.-Q.; Cheng, Y.; Xiao, W.-J. Photoinduced Cobalt-Catalyzed Desymmetrization of Dialdehydes to Access Axial Chirality. *J. Am. Chem. Soc.* **2023**, *145*, 6944−6952.
(10) Wang, Y.; Mi, R.; Yu, S.; Li, X. Expedient synthesis of axially and centrally chiral diaryl ethers via cobalt-catalyzed photoreductive desymmetrization. *ACS Catal.* **2024**, *14*, 4638−4647.
(11) Zhang, F.-H.; Guo, X.; Zeng, X.; Wang, Z. Asymmetric 1,4-functionalization of 1,3-enynes via dual photoredox and chromium catalysis. *Nat. Commun.* **2022**, *13*, 5036.
(12) (a) *Modern allene chemistry*; Krause, N., Hashmi, A. S. K., Eds.; Wiley VCH: Weinhein, 2004. (b) Yu, S.; Ma, S. Allenes in catalytic asymmetric synthesis and natural product syntheses. *Angew. Chem., Int. Ed.* **2012**, *51*, 3074−3112.
(13) (a) Ye, J.; Ma, S. Conquering three-carbon axial chirality of allenes. *Org. Chem. Front.* **2014**, *1*, 1210−1224. (b) Chu, W.-D.; Zhang, Y.; Wang, J. Recent advances in catalytic asymmetric synthesis of allenes. *Catal. Sci. Technol.* **2017**, *7*, 4570−4579. (c) Huang, X.; Ma, S. Allenation of terminal alkynes with aldehydes and ketones. *Acc. Chem. Res.* **2019**, *52*, 1301−1312. (d) Neff, R. K.; Frantz, D. E. Recent advances in the catalytic syntheses of allenes: a critical assessment. *ACS Catal.* **2014**, *4*, 519−528.
(14) (a) Dherbassy, Q.; Manna, S.; Talbot, F. J. T.; Prasitwatcharakorn, W.; Perry, G. J. P.; Procter, D. J. Copper-catalyzed functionalization of enynes. *Chem. Sci.* **2020**, *11*, 11380−11393. (b) Fu, L.; Greßies, S.; Chen, P.; Liu, G. Recent advances and perspectives in transition metal-catalyzed 1,4-functionalizations of unactivated 1,3-enynes for the synthesis of allenes. *Chin. J. Chem.* **2020**, *38*, 91−100.
(15) Li, S.; Yuan, K.; Zhang, G.; Guo, R. Recent advances in the synthesis of chiral allenes via asymmetric 1,4-difunctionalization of 1,3-enynes. *Eur. J. Org Chem.* **2024**, *27*, No. e202301316.
(16) (a) Liao, Y.; Yin, X.; Wang, X.; Yu, W.; Fang, D.; Hu, L.; Wang, M.; Liao, J. Enantioselective synthesis of multisubstituted allenes by cooperative Cu/Pd-catalyzed 1,4-arylboration of 1,3-enynes. *Angew. Chem., Int. Ed.* **2020**, *59*, 1176−1180. (b) Zheng, W.-F.; Zhang, W.; Huang, C.; Wu, P.; Qian, H.; Wang, L.; Guo, Y.-L.; Ma, S. Tetrasubstituted allenes via the palladium-catalysed kinetic resolution of propargylic alcohols using a supporting ligand. *Nat. Catal.* **2019**, *2*, 997−1005.
(17) (a) Zhang, W.; Zheng, S.; Liu, N.; Werness, J. B.; Guzei, I. A.; Tang, W. Enantioselective bromolactonization of conjugated (Z)-enynes. *J. Am. Chem. Soc.* **2010**, *132*, 3664−3665. (b) Li, X.; Sun, J. Organocatalytic enantioselective Synthesis of chiral allenes: remote asymmetric 1,8-Addition of indole imine methides. *Angew. Chem., Int. Ed.* **2020**, *59*, 17049−17054. (c) Qian, D.; Wu, L.; Lin, Z.; Sun, J. Organocatalytic synthesis of chiral tetrasubstituted allenes from racemic propargylic alcohols. *Nat. Commun.* **2017**, *8*, 567−575.
(18) (a) Wang, F.; Wang, D.; Zhou, Y.; Liang, L.; Lu, R.; Chen, P.; Lin, Z.; Liu, G. Divergent synthesis of CF_3-substituted allenyl nitriles by ligand-controlled radical 1,2- and 1,4-addition to 1,3-enynes. *Angew. Chem., Int. Ed.* **2018**, *57*, 7140−7145. (b) Zhu, X.; Deng, W.; Chiou, M.-F.; Ye, C.; Jian, W.; Zeng, Y.; Jiao, Y.; Ge, L.; Li, Y.; Zhang, X.; Bao, H. Copper-catalyzed radical 1,4-difunctionalization of 1,3-enynes with alkyl diacyl peroxides and N-fluorobenzenesulfonimide. *J. Am. Chem. Soc.* **2019**, *141*, 548−559. (c) Song, Y.; Song, S.; Duan, X.; Wu, X.; Jiang, F.; Zhang, Y.; Fan, J.; Huang, X.; Fu, C.; Ma, S. Copper-catalyzed radical approach to allenyl iodides. *Chem. Commun.* **2019**, *55*, 11774−11777. (d) Zhang, K.-F.; Bian, K.-J.; Li, C.; Sheng, J.; Li, Y.; Wang, X.-S. Nickel-catalyzed carbofluoroalkylation of 1,3-enynes to access structurally diverse fluoroalkylated allenes. *Angew. Chem., Int. Ed.* **2019**, *58*, 5069−5074. (e) Song, Y.; Fu, C.; Ma, S. Copper-catalyzed syntheses of multiple functionalizatized allenes via three-component reaction of enynes. *ACS Catal.* **2021**, *11*, 10007−10013. (f) Li, Y.; Bao, H. Radical transformations for allene synthesis. *Chem. Sci.* **2022**, *13*, 8491−8506. (g) Li, X.; Li, N.; Yang, L.; Jiang, R.; He, Y.; Shi, J.; Jiang, Z.-X.; Yang, Z. Switchable nucleophilic site enables expedient synthesis of CF_3-containing thiazoles and allenes from 1,3-enynes. *ACS Catal.* **2023**, *13*, 12755−12765. (h) Shan, Q.-C.; Zhao, Y.; Wang, S.-T.; Liu, H.-F.; Duan, X.-H.; Guo, L.-N. Nickel-catalyzed modular four-component 1,4-alkylcarbonylation of 1,3-enynes to tetra-substituted CF_3−allenyl ketones. *ACS Catal.* **2024**, *14*, 2144−2150.
(19) Hansmann, M. M.; Melaimi, M.; Bertrand, G. Crystalline monomeric allenyl/propargyl radical. *J. Am. Chem. Soc.* **2017**, *139*, 15620−15623.
(20) Zeng, Y.; Chiou, M.-F.; Zhu, X.; Cao, J.; Lv, D.; Jian, W.; Li, Y.; Zhang, X.; Bao, H. Copper-catalyzed enantioselective radical 1,4-difunctionalization of 1,3-enynes. *J. Am. Chem. Soc.* **2020**, *142*, 18014−18021.
(21) Dong, X.-Y.; Zhan, T.-Y.; Jiang, S.-P.; Liu, X.-D.; Ye, L.; Li, Z.-L.; Gu, Q.-S.; Liu, X.-Y. Copper-catalyzed asymmetric coupling of allenyl radicals with terminal alkynes to access tetrasubstituted allenes. *Angew. Chem., Int. Ed.* **2021**, *60*, 2160−2164.
(22) (a) Chen, Y.; Wang, J.; Lu, Y. Decarboxylative 1,4-carbocyanation of 1,3-enynes to access tetra-substituted allenes via copper/photoredox dual catalysis. *Chem. Sci.* **2021**, *12*, 11316−11321. (b) Chen, Y.; Zhu, K.; Huang, Q.; Lu, Y. Regiodivergent sulfonylarylation of 1,3-enynes via nickel/photoredox dual catalysis. *Chem. Sci.* **2021**, *12*, 13564−13571.
(23) For selected examples with nickel catalysis, see (a) Xu, T.; Wu, S.; Zhang, Q.-N.; Wu, Y.; Hu, M.; Li, J.-H. Dual photoredox/nickel-catalyzed 1,4-sulfonylarylation of 1,3-enynes with sulfinate salts and aryl halides: entry into tetrasubstituted allenes. *Org. Lett.* **2021**, *23*, 8455−8459. (b) Li, C.; Hu, D.-D.; Jin, R.-X.; Wu, B.-B.; Wang, C.-Y.; Ke, Z.; Wang, X.-S. Selective 1,4-arylsulfonation of 1,3-enynes via photoredox/nickel dual catalysis. *Org. Chem. Front.* **2022**, *9*, 788−794. (c) Liu, W.; Liu, C.; Wang, M.; Kong, W. Modular synthesis of multifunctionalized CF_3-allenes through selective activation of saturated hydrocarbons. *ACS Catal.* **2022**, *12*, 10207−10221. (d) Cheng, C.; Lv, G.-F.; Wu, S.; Li, Y.; Li, J.-H. Nickel photocatalyzed reductive 1,4-dicarbofunctionalization of 1,3-enynes with N-methylamines and organohalides enabled by site-selective C(sp^3)−H functionalization. *Org. Lett.* **2023**, *25*, 4236−4240. (e) Wang, Q.; Chen, Y.; Peng, K.; Li, Y.; Cheng, L.; Deng, G.-J. Three-component cross-electrophile 1,4-alkylarylation of 1,3-enynes by merging nickel and photoredox catalysis. *Org. Lett.* **2023**, *25*, 8889−8894. (f) Jeon, J. H.; Kim, G. H.; Lee, H. S.; Kim, D. H.; Lee, S.; Choe, W.; Jung, B.; Rohde, J.-U.; Hong, S. Y. Chemo- and

https://doi.org/10.1021/acscatal.4c04330
ACS Catal. 2024, 14, 14048−14057

regioselective nickel-catalyzed reductive 1,4-alkylarylation of 1,3-enynes through an L_2NiAr intermediate. *ACS Catal.* **2024**, *14*, 8996−9007. (g) Chen, H.; Zhu, C.; Yue, H.; Rueping, M. Group 14 elements hetero-difunctionalizations via nickel-catalyzed electroreductive cross-coupling. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202306498. (h) Zhu, C.; Chen, H.; Yue, H.; Rueping, M. Electrochemical chemo- and regioselective arylalkylation, dialkylation and hydro(deutero)alkylation of 1,3-enynes. *Nat. Synth.* **2023**, *2*, 1068−1081.

(24) For selected examples with copper catalysis, see (a) Sun, Q.; Zhang, X.-P.; Duan, X.; Qin, L.-Z.; Yuan, X.; Wu, M.-Y.; Liu, J.; Zhu, S.-S.; Qiu, J.-K.; Guo, K. Photoinduced merging with copper- or nickel-catalyzed 1,4-cyanoalkylarylation of 1,3-enynes to access multiple functionalizatized allenes in batch and continuous flow. *Chin. J. Chem.* **2022**, *40*, 1537−1545. (b) Yang, L.; Zeng, L.; Zhu, Y.; Wang, Q.; Li, J. Copper-catalyzed photoredox 1,4-amidocyanation of 1,3-enynes with N-amidopyridin-1-ium salts and TMSCN: Facile access to α-amido allenyl nitriles. *Chin. Chem. Lett.* **2024**, *35*, 109685. (c) Pu, Y.; Ding, S.; Zhao, H.; Xue, Q.; Zhang, H.; Xie, X.; Shang, Y.; Wang, J. Three-component synthesis of multiple functionalized allenes via copper/photoredox dual catalyzed 1,4-alkylcyanation of 1,3-enynes. *Org. Lett.* **2024**, *26*, 1834−1839. (d) Wang, L.; Qin, W. Copper-initiated regiodivergent chloropentafluorosulfanylation of 1,3-enynes under substrate control. *Org. Lett.* **2024**, *26*, 5049−5054. (e) Chen, Y.; Wanga, Q.; Penga, K.; Chenga, L.; Shea, Z.; Deng, G.-J. Copper/photoredox-catalyzed regioselective 1, 4-addition to 1,3-enynes: a convenient access to perfluoroalkylated allenes. *Chin. J. Chem.* **2024**, *42*, 2023−2028.

(25) For selected examples with NHC catalysis, see (a) Chen, L.; Lin, C.; Zhang, S.; Zhang, X.; Zhang, J.; Xing, L.; Guo, Y.; Feng, J.; Gao, J.; Du, D. 1,4-Alkylcarbonylation of 1,3-enynes to access tetra-substituted allenyl ketones via an NHC-catalyzed Radical relay. *ACS Catal.* **2021**, *11*, 13363−13373. (b) Cai, Y.; Chen, J.; Huang, Y. N. N-Heterocyclic Carbene-Catalyzed 1,4-Alkylacylation of 1,3-Enynes. *Org. Lett.* **2021**, *23*, 9251−9255. (c) Wang, L.; Ma, R.; Sun, J.; Zheng, G.; Zhang, Q. NHC and visible light-mediated photoredox co-catalyzed 1,4-sulfonylacylation of 1,3-enynes for tetrasubstituted allenyl ketones. *Chem. Sci.* **2022**, *13*, 3169−3175.

(26) (a) Dai, L.; Zhou, X.; Guo, J.; Dai, X.; Huang, Q.; Lu, Y. Diastereo- and atroposelective synthesis of N-arylpyrroles enabled by light-induced phosphoric acid catalysis. *Nat. Commun.* **2023**, *14*, 4813. (b) Dai, L.; Guo, J.; Huang, Q.; Lu, Y. Asymmetric multifunctionalization of alkynes via photo-irradiated organocatalysis. *Sci. Adv.* **2022**, *8*, No. eadd2574. (c) Zhou, X.; Huang, Q.; Guo, J.; Dai, L.; Lu, Y. Enantioselective De Novo synthesis of α,α-diaryl ketones from alkynes. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202310078.

(27) (a) Mei, G.-J.; Koay, W. L.; Guan, C.-Y.; Lu, Y. Atropisomers beyond the C−C axial chirality: advances in catalytic asymmetric synthesis. *Chem* **2022**, *8*, 1855−1893. (b) Mei, G.-J.; Wong, J. J.; Zheng, W.; Nangia, A. A.; Houk, K. N.; Lu, Y. Rational design and atroposelective synthesis of N−N axially chiral compounds. *Chem* **2021**, *7*, 2743−2757. (c) Xiang Alvin Tan, C.; Li, R.; Zhang, F.; Dai, L.; Ullah, N.; Lu, Y. Synthesis of axially chiral CF_3-substituted 2-arylpyrroles by sequential phosphine-catalyzed asymmetric [3 + 2] annulation and oxidative central-to-axial chirality transfer. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202209494.

(28) Wang, D.; Zhu, N.; Chen, P.; Lin, Z.; Liu, G. Enantioselective decarboxylative cyanation employing cooperative photoredox catalysis and copper catalysis. *J. Am. Chem. Soc.* **2017**, *139*, 15632−15635.

(29) (a) Zhang, W.; Wang, F.; McCann, S. D.; Wang, D.; Chen, P.; Stahl, S. S.; Liu, G. Enantioselective cyanation of benzylic C-H bonds via copper-catalyzed radical relay. *Science* **2016**, *353*, 1014−1018. (b) Li, J.; Zhang, Z.; Wu, L.; Zhang, W.; Chen, P.; Lin, Z.; Liu, G. Site-specific allylic C−H bond functionalization with a copper-bound N-centred radical. *Nature* **2019**, *574*, 516−521.

(30) (a) Zarghi, A.; Rao, P. N. P.; Knaus, E. E. Sulfonamido, azidosulfonyl and N-acetylsulfonamido analogues of rofecoxib: 4-[4-(N-acetylsulfonamido)phenyl]-3-(4-methanesulfonylphenyl)-2(5H)-furanone is a potent and selective cyclooxygenase-2 inhibitor. *Bioorg. Med. Chem. Lett.* **2004**, *35*, 1957−1960. (b) Feng, Z.; Chu, F.; Guo, Z.; Sun, P. Synthesis and anti-inflammatory activity of the major metabolites of imrecoxib. *Bioorg. Med. Chem. Lett.* **2009**, *19*, 2270−2272.

(31) (a) Lu, R.; Yang, T.; Chen, X.; Fan, W.; Chen, P.; Lin, Z.; Liu, G. Enantioselective copper-catalyzed radical cyanation of propargylic C−H bonds: easy access to chiral allenyl nitriles. *J. Am. Chem. Soc.* **2021**, *143*, 14451−14457. (b) Cheng, Z.; Yang, T.; Li, C.; Deng, Y.; Zhang, F.; Chen, P.; Lin, Z.; Ma, S.; Liu, G. Site-selective sp^2 C−H cyanation of allenes via copper-catalyzed radical relay. *J. Am. Chem. Soc.* **2023**, *145*, 25995−26002.

(32) Okada, K.; Okamoto, K.; Oda, M. A new and practical method of decarboxylation: photosensitized decarboxylation of N-acyloxyphthalimides via electron-transfer mechanism. *J. Am. Chem. Soc.* **1988**, *110*, 8736−8738.

(33) Prier, C. K.; Rankic, D. A.; MacMillan, D. W. C. Visible light photoredox catalysis with transition metal complexes: applications in organic synthesis. *Chem. Rev.* **2013**, *113*, 5322−5363.

https://doi.org/10.1021/acscatal.4c04330
ACS Catal. 2024, 14, 14048−14057